ADAM LISTER | CV

adamlisteruk@proton.me • adam-lister.com

CURRENT

Position Scientist I, Univeristy of Wisconsin – Madison

Bio Leader in New Physics Searches, Neutrino Oscillations, Detector Physics, and Detector Calibration. Collaborator on two international neutrino experiments, NOvA (200 members) and DUNE (1600 members)

Leadership NOvA NuX (New Physics in Oscillations) Convener

Projects *NOvA Sterile Neutrino Search* • Planning and executing search for sterile neutrinos using neutrino and antineutrino mode data from the NuMI beam.

BNB Neutrinos at NOvA • Leader in understanding neutrinos from NOvA's second beamline, including flux simulation, integration into NOvA software, and developing data quality requirements.

OSCAr Test Stand • Currently working to develop test stand to measure never-before-measured transverse component of diffusion of ionisation electrons in LArTPCs. First data taken December 2025.

EDUCATION AND EMPLOYMENT

2019-2024 **Postdoctoral Research Associate** *University of Wisconsin – Madison, USA*
Collaborator on the NOvA and DUNE experiments

2015-2019 **PhD Experimental Particle Physics** *Lancaster University, UK*
Collaborator on the MicroBooNE experiment
Thesis: Constraint of Systematic Uncertainties in an Electron Neutrino Search Using Muon Neutrinos at MicroBooNE

2013-2014 **MSc Particles, Strings, and Cosmology** *Durham University, UK*
Thesis: Neutrino Oscillations in Long-Baseline Experiments

2010-2013 **BSc Astrophysics (1st class)** *Aberystwyth University, UK*

LEADERSHIP

2025-2026 **NOvA Executive Committee** (elected)
Elected to advise the spokespeople on science and management.

2024-present **NOvA NuX Group Convener** (appointed) ∼30 people
Managing searches for New Physics in neutrino oscillations • Two PRL publications during tenure, with another in internal review • Seven active analyses, each targeting high-impact journals.

2022-2024 **NOvA Detector Systematics Group Convener** (appointed) ∼30 people
Delivered simulation improvements (Geant4, neutron model, detector simulation) • Updated calibration scheme which increased speed and reduced failure rates.

2020-2022 **NOvA Production Group Convener** (appointed) ∼5 people
Delivered largest ever file production used for all recent NOvA publications • Developed computing infrastructure & monitoring tools • Developed new storage-informed production model, enabling backprocessing of ∼ 1 PB of cosmic-ray data which was used in NOvA's flagship neutrino oscillation search.

AWARDS, OUTREACH & SERVICE

Awards Contributor to successful UW-Madison DOE Base Grants (’20,’23,’26) • $10,000 Fermilab Neutrino Physics Center Fellowship (’18) • Lancaster University Department Prize (’19)

Outreach NOvA public website manager (’24-present) • NOvA social media manager (’24-present) • Fermilab LGBTQ+ group Pride Parade (’22,’24) • NOvA Experiment Facility Tour Guide (’23-’24) • Coordinated and featured in articles for APS Physics Magazine and Physics World for NOvA Sterile Neutrino Results • Fermilab Open House volunteer (’17,’19)

Service NOvA Paper Review Committee for neutrino interactions measurement (’26) • NOvA Analysis Review Committee for cosmic-ray muon measurement (’23-’25) • Session chair at Lepton-Photon 2025 • NOvA Workshop organiser (’24,’25) • DUNE Plot Style Committee (’22-’23)

PRESENTATIONS

Jun 2026 **Neutrino 2026** *Poster*
Future Sterile Neutrino Searches in NOvA using the Booster Neutrino Beam

Jan 2026 **NuPhys 2026** *Contributed Talk*
NOvA’s Current and Future Sterile Neutrino Searches

Aug 2025 **Lepton-Photon 2025** *Contributed Talk*
Beyond Standard Model Neutrino Oscillation Results from NOvA

Jun 2025 **WIN 2025** *Contributed Talk*
Beyond Standard Model Neutrino Oscillation Results from NOvA

Apr 2025 **U. Chicago, U. Cincinnati, U. Mississippi** *Invited Colloquia/Seminars*
Looking for Sterile Neutrinos with the NOvA Experiment

Jun 2024 **Neutrino 2024** *Poster*
Improving NOvA’s Sterile Neutrino Search with the Booster Neutrino Beam

Dec 2023 **NuPhys 2023** *Invited Talk*
Neutrino Oscillations at NOvA

Sep 2023 **Cornell University** *Invited Seminar*
Searching For Sterile Neutrinos With The NOvA Experiment

Dec 2022 **CPAD Workshop 2022** *Contributed Talk*
Effects of Ionisation Electron Diffusion on Calibrations in LArTPCs

Jun 2022 **55th Annual Fermilab Users’ Meeting** *Invited Talk*
Probing Neutrino Oscillations at The NOvA Experiment

Jun 2020 **Neutrino 2020** *Poster*
Event Selection and Systematics for the NOvA Sterile Neutrino Search

Dec 2018 **University of Wisconsin – Madison** *Invited Seminar*
MicroBooNE and the Path to Resolving the MiniBooNE Low-Energy Excess

Mar 2018 **IOP HEPP and APP Meeting** *Contributed Talk*
Measuring Electron Diffusion in MicroBooNE

Jun 2017 **50th Annual Fermilab Users’ Meeting** *Poster*
Towards a Longitudinal Electron Diffusion Measurement in MicroBooNE

Jul 2016 **Neutrino 2016** *Poster*
Measurements of ν_μ CC Interactions by the MicroBooNE Experiment

SELECT PUBLICATIONS

Listed below are those publications where I have had a significant contribution. A full list of publications can be found on ORCiD, or on iNSPIRE.

2026 **NOvA's Current and Future Sterile Neutrino Searches**
Adam Lister, for the NOvA Collaboration, arXiv 2602.11345

2026 **Precision measurement of neutrino oscillation parameters with 10 years of data from the NOvA experiment**
The NOvA Collaboration, PRL 136 011802

2026 **Explanation of the Seasonal Variation of Cosmic Multiple Muon Events Observed with the NOvA Near Detector**
The NOvA Collaboration, PRD 113 012001

2025 **Dual-Baseline Search for Active-to-Sterile Neutrino Oscillations in NOvA**
The NOvA Collaboration, PRL 134 081804

2024 **Search for CP-Violating Neutrino Nonstandard Interactions with the NOvA Experiment**
The NOvA Collaboration, PRL 133 201802

2022 **Investigations on a fuzzy process: effect of diffusion on calibration and particle identification in Liquid Argon Time Projection Chambers**
A. Lister & M. Stancari, JINST 17 P07016

2022 **Search for an Excess of Electron Neutrino Interactions in MicroBooNE Using Multiple Final-State Topologies**
The MicroBooNE Collaboration, PRL 128 241801

2022 **Search for an anomalous excess of charged-current ν_e interactions without pions in the final state with the MicroBooNE experiment**
The MicroBooNe Collaboration, PRD 105 112004

2021 **Measurement of the longitudinal diffusion of ionization electrons in the MicroBooNE detector**
The MicroBooNE Collaboration, JINST 16 P09025

2020 **Measurement of differential cross sections for muon neutrino-argon charged-current interactions with protons and no pions in the final state in the MicroBooNE detector**
The MicroBooNE Collaboration, PRD 102 112013

2020 **Deep Underground Neutrino Experiment (DUNE), Far Detector Technical Design Report, Volume IV: Far Detector Single-phase Technology**
The DUNE Collaboration, JINST 15 T08010